Kyle Wodehouse

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Education

B.ChE. in Chemical Engineering, University of Delaware 09/2023 - 05/2027
Minor: Mathematics and Data Science | GPA: 4.00/4.00
Thesis Advisor: Srikanth Pilla | Honors thesis focus: Thermal transport in semicrystalline polymers

Research and Industry Experience

R&D Intern/Co-op 06/2024 - present
DuPont Water Solutions | Wilmington, DE

- Characterized ion exchange resin properties and evaluated product performance in application-specific tests.
- Conducted free radical polymerization reactions and sampled in situ to evaluate conversion and gel points for raw material replacement and additive selection.
- Analyzed sugar hydrogenation reaction samples with high-performance liquid chromatography and automated signal processing in Python.
- Built Python/OpenCV image-analysis software to measure resin bead diameters from microscopy images and summarize particle-size distributions.
- Designed and executed experiments evaluating novel PFAS sorbents for ultra-short-chain contaminated streams.
- Developed enzyme immobilization and assay procedures for evaluating polymer supports for commercially relavent enzymes.
- Built automated experimental setup with abandoned lab equipment controlled over serial communications.

Undergraduate Researcher, Jayaraman Lab 02/2025 - present
Department of Chemical and Biomolecular Engineering | University of Delaware, Newark, DE

- Investigated thermal transport in polymers using coarse-grained molecular dynamics (MD) models that separate crystalline fraction from crystallite size, shape, and orientation.
- Measured directional thermal conductivity with nonequilibrium molecular dynamics simulations in LAMMPS.
- Developed high-throughput Python workflows to generate morphology-controlled simulation inputs, submit jobs to high-performance computing resources, and analyze LAMMPS outputs.
- Communicated research progress and technical findings through monthly update slideshows.

Undergraduate Researcher, Hernandez-Pagan Lab 08/2023 - 05/2024
Department of Chemistry | University of Delaware, Newark, DE

- Expanded a procedure for doping In_2O_3 nanoparticles with transition metals, including Mg, Mn, and Y.
- Synthesized indium oxide nanoparticles using a Schlenk line under vacuum and nitrogen atmosphere and characterized them with UV-vis spectroscopy and X-ray diffraction.
- Developed a procedure for growing noble metal crystals on indium oxide nanoparticles.

Selected Honors and Awards

Future Leaders in Chemical Engineering Symposium, NC State University	2026
Sandra & J Michael Bowman Scholarship	2026
Slocomb Engineering Scholarship	2026
Tau Beta Pi Scholarship	2026
American Chemical Society Award in Chemical Engineering	2026
Barry M. Goldwater Scholarship	2026
James F. Kearns Scholarship in Chemical Engineering	2025 & 2026
John R. Eagle Scholarship	2025 & 2026
Tau Beta Pi Engineering Honor Society	2024
Industrial Sponsors First Year Scholarship	2024
Chick-fil-a Remarkable Futures Scholarship	2024

Patents and Patent Applications

1. Lorzing, G.; Sengupta, S.; Trejo, J.; Martin, C.; **Wodehouse, K.** "Catalyst and Process for the Preparation of Sugar Alcohols." *International Patent Application* WO 2026/096843, published May 7, 2026.

Presentations

Poster Presentations

2. **Wodehouse, K.**; Lorzing, G.; Croft, J.; Tomey, K.; Sengupta, S. "Ruthenium Supported Ion Exchange Resin as a Glucose Hydrogenation Catalyst." Stanford Research Conference, Stanford University, April 2026.

1. **Wodehouse, K.**; Lorzing, G.; Sengupta, S. "Ruthenium Supported Ion Exchange Resin as a Glucose Hydrogenation Catalyst." Winter Research Symposium, University of Delaware, February 2026.

Oral Presentations

3. **Wodehouse, K.** "Ruthenium Supported Ion Exchange Resin as a Glucose Hydrogenation Catalyst." Gulf Coast Undergraduate Research Symposium, Rice University, October 2025.
2. **Wodehouse, K.** "Measuring Ion Exchange Resin Beads via Optical Microscopy." DuPont Water Solutions, Wilmington, DE, May 2025.
1. **Wodehouse, K.**; Lorzing, G. "Analyzing Sugar Samples From Ruthenium Catalyst." DuPont Water Solutions, Wilmington, DE, August 2024.

Teaching Experience

Grader, CHEG332: Chemical Engineering Kinetics	Fall 2026
Undergraduate Teaching Assistant, CHEM444: Physical Chemistry II	Spring 2026
Grader, CHEG325: Chemical Engineering Thermodynamics II	Spring 2026
Grader, CHEG112: Introduction to Chemical Engineering	Spring 2025

Technical Skills

Programming	Python (NumPy, pandas, SciPy, OpenCV, PyTorch), Git/GitHub, Bash
Molecular Simulation	LAMMPS, coarse-grained MD, nonequilibrium MD, HPC job workflows, SLURM
Characterization	HPLC, UV-Vis spectroscopy, X-ray diffraction, IR spectroscopy, optical microscopy, SEM, TEM, SEM-EDX, AFM-IR, ToF-SIMS, ICP-OES
Laboratory Techniques	Free radical polymerization, copolymer functionalization, titration (Mohr, Volhard, Fajans), Bradford assay, Schlenk line synthesis, nanoparticle synthesis, parallel plate rheometry, cone and plate rheometry

Leadership and Activities

Academic Chair Committee Chair, American Institute of Chemical Engineers	05/2026 - present
President & Founding Member, Chem-E Jeopardy	08/2025 - present
Recruitment Committee, American Institute of Chemical Engineers	08/2023 - 05/2026
Member, Chem-E Cube	08/2023 - present
Instrumental Music, Cello	08/2023 - present

UD Cello Studio / UD Chamber Ensembles / UD Chamber Orchestra / Berks Sinfonietta

Additional activities: Blue Hen Cornhole (02/2024 - present), Academic Competition Club (08/2023 - 05/2024)